MARK IMADE

markimade01@gmail.com | linkedin.com/in/mark-imade | github.com/qlemon-56

Education

University of York Expected June 2028
B.Eng. in Electronic and Computer Engineering
- Relevant Modules: Programming and Digital Interfacing, Digital Electronics, Analog Electronics

University of York International Pathway College Jan 2025 – Aug 2025
Foundation Certificate in Computer Science and Engineering
- Modules: Electronics, Robotics, Advanced Mathematics
- Grade: 95% Overall

Experience

Full-Stack Software Developer – York Community Consulting Mar 2026 – Present
- Develop and maintain YCC's website, implementing new features to support project delivery
- Implemented RESTful APIs to serve application data, integrating Redis caching to reduce latency
- Refactored fragmented search filters into a unified and reusable TypeScript component, reducing UI clutter and improving application review process
- Researched and implemented backup strategy in Google Firebase to ensure business continuity
- Conducted code reviews and carried out QA tests to maintain code quality and consistency

Electronics & Chassis Team Member – York Formula Student Sept 2025 – Present
- Contributed to the development of a battery powered race car for YFS's FSUK debut
- Prototyped shared mount for diagnostic test points and 3 power components in Autodesk Fusion
- Supported development by taking measurements, machining, and polishing vehicle parts

IEUK Technology and Engineering Intern – Bright Network June 2025
- Analysed 400k+ lines of web traffic logs using Python, accurately identifying suspicious activity patterns and recommending effective bot control strategies in a technical report
- Gained exposure to technology and engineering industry practices through presentations and Q&A sessions

Projects

Arvid: Game & Handheld Console | C++, Typescript, HTML, CSS, STM32 Feb 2026 – June 2026
- Collaborated with 5 students to develop a game console and 2D game from scratch using C++
- Programmed game logic using an ECS design pattern enabling quick prototyping
- Aided design of custom PCB and reviewed console prototypes created in Autodesk Fusion
- Integrated joystick component by mapping analogue voltage readings to 6 distinct states
- Optimised game rendering by implementing display caching
- Developed a promotional website using Next.js, TypeScript, Tailwind CSS based on Figma designs

Seven Segment Display Library | C++, STM32 Dec 2025 – Jan 2026
- Developed a C++ library for STM32 microcontrollers to control 7-segment displays with customisable refresh rates and expandable character set
- Designed a multi-threaded control architecture, implementing mutex-based locks to eliminate race conditions during concurrent display tasks

Skills

Languages: C++, Python, JavaScript, TypeScript, HTML, CSS
Technologies: Git, React, Next.js, Tailwind CSS, Firebase, Redis
Software: Autodesk Fusion, Figma, MS Excel, MS Word, AMD Vivado